
Education

University of Management and Technology

Bachelor of Science in Software Engineering. GPA: 3.7/4.0

Lahore, Pakistan

Expected Graduation: February, 2027

Relevant Coursework: Machine Learning, Natural Language Processing, Data Mining, Data Science, Data Structures & Algorithms, Database Systems, Probability & Statistics

Queen Mary College

Intermediate Pre-Engineering

Lahore, Pakistan

2020-2022

Grade: A

Projects

1. AI Job Market Analysis & Salary Prediction

- Analyzed 15,000 AI/ML job postings to identify skill demand trends and salary patterns.
- Applied K-Means clustering to discover job role groups based on required skills.
- Built a Random Forest regression model ($R^2 \approx 0.58$) to predict salaries and visualize insights.

2. Django User Authentication & Profile Management Web Application

- Engineered a secure authentication and profile management system using Django's built-in authentication framework.
- Implemented user registration, login/logout, and session-based access control.
- Designed a custom Profile model (One-to-One with User) for extended user data.
- Developed profile update functionality with image upload and server-side validation.
- Applied template inheritance and modular project organization for maintainable code.

3. Desktop Application – PyQt5 & Qt Designer

- Developed a GUI-based desktop application using Qt Designer and Python (PyQt).
- Implemented event-driven programming with modular UI and business logic.
- Integrated persistent data storage using files/databases.
- Applied object-oriented programming principles for maintainable code.

4. Dynamic CV Builder – HTML, CSS, JavaScript

- Built a client-side web application for generating CVs through form-based input.
- Implemented real-time DOM manipulation for dynamic CV rendering.
- Designed responsive layouts using HTML5 and CSS3.
- Applied client-side validation for structured data handling.

5. Portfolio Website

- Developed a responsive portfolio website using HTML, CSS, and JavaScript to showcase projects and technical skills.
- Optimized the layout for desktop and mobile devices using responsive design principles.

6. NLP Emotion Detection System – Python, DistilBERT, Hugging Face

- Developed a multi-class emotion detection system using the DistilBERT transformer model.
- Fine-tuned a pre-trained DistilBERT model using Hugging Face Transformers and PyTorch.
- Performed text preprocessing, tokenization, and dataset preparation.
- Evaluated model performance using accuracy, precision, recall, and F1-score.

Certificates

1. Machine Learning with Python – freeCodeCamp

- Learned supervised learning, regression, classification, and model evaluation using Python and Scikit-learn.
- Applied data preprocessing, feature engineering, and model training through hands-on projects.

2. Project Management Fundamentals – Microsoft | Coursera

- Learned project planning, scheduling, risk management, stakeholder communication, and project lifecycle fundamentals.
- Applied project management principles and best practices through practical learning activities.

3. Responsive Web Design – freeCodeCamp

- Learned HTML5, CSS3, and responsive web design principles.
- Built 20+ practice projects to strengthen front-end development skills.

Technical skills

- **Programming Languages:** Python, Java, C++, JavaScript, SQL
- **Web Technologies:** HTML5, CSS3, Django, React, Node.js
- **Machine Learning & Tools:** Pandas, NumPy, Scikit-learn, PyTorch, Hugging Face Transformers, Matplotlib, Seaborn, PyQt5, Selenium
- **Version Control & Development Tools:** Git, GitHub, VS Code, Kali Linux, Cisco Packet Tracer